



Berna Hanedar Assoc. Prof. Dr.



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About me

Nationality : Turkish

Date of the doctorate : 04/01/2017

Master's end dates : 1/06/2010

Master's starting date : 15/09/2008

Bachelor's end dates : 5/07/2008

Bachelor's starting date : 5/09/2004

Stays abroad :

USA, 27/07/2022-15/02/2023

USA, 19/10/2014-01/02/2016

Germany, 15/09/2009-15/09/2010

Research position : Kırklareli University, 15/09/2010- Cont.

Over 11 years of working experience on materials engineering and research. Extensive knowledge of piezoelectric properties and perovskite structures. Deep hands-on experience on DFT and MD simulation calculations. New physical and chemical studies of Nanomaterials, 2D-Materials and MXenes. I have reviewed of many article for prior journals like; 2D materials, ACS Omega, Nanoscale, IOP Science, Royal Society of Chemistry, etc.

Skills

DFT Simulations

Writing Skill

English

(*)The skill scale is from 0 (Fundamental)

Academic Experience

2023-cont.	Postdoctoral Researcher,	Koc University
2021-cont.	Associate Professor Doctor	Kırklareli University
2016-2021	Research Assistant Doctor	Kırklareli University
2010-2016	Research Assistant	Kırklareli University

Educational Experience

July2022-Feb2023	Rice University, Houston, Texas, USA http://biygroup.blogs.rice.edu/members	Visiting Researcher
Jau2019-Feb2019	Bilkent University UNAM, Ankara, TURKEY http://unam.bilkent.edu.tr/durgun/	Visiting Researcher
Jun2018-Aug2018	Izmir Institute of Technology, Izmir, TURKEY http://sahingroup.iyte.edu.tr	Visiting Researcher
Oct2014-Feb2016	Texas AM University, Texas, USA Supported by TUBITAK and Council of Higher Education	Visiting Scholar
Sep2009-Sep2010	Universitat Augsburg, Augsburg, GERMANY Supported by (ERASMUS)	Erasmus Student

References

Boris Yakobson	Prof. Dr. (biy@rice.edu) <i>Rice University, Materials Science and NanoEngineering Department</i>	Post.Doc.advisor
Tahir Cagin	Prof. Dr. (tcagin@tamu.edu) <i>Texas AM University, Material Science and Engineering Department</i>	Ph.D. Coadvisor
Ulrich Eckern	Prof. Dr. (ulrich.eckern@physik.uni-augsburg.de) <i>Augsburg University, Physics Department</i>	M. Sc. Coadvisor

Education

2011-2016	Ph.D. in Physics <i>Thesis Title: A study of the relation of piezoelectric properties and nano structures through methods in computational physics.</i>	Yildiz Technical University
2008-2010	M.Sc. in Physics <i>Thesis Title: DFT based calculations of the electronic structure and transmission coefficient of nanoscaled molecular contacts</i>	Sakarya University
2004-2008	B.Sc. in Physics	Kocaeli University

Teaching Experience

General Physics I/II, Thermodynamics, Quantum Physics, Modern Physics, Statistical Physics, Condensed Matter Physics I/II, Atomic Physics, Computer Programming in Physics, Material Science, Scientific English.

Research Interest

My research focuses on developing and applying the tools of theoretical and computational physics for the proper understanding of the physical properties of objects in the nanoscale size range. I have address critical challenges in various areas including 2D materials, surface phenomena, nanostructures, magnetism, spintronics, and hydrogen storage from nano to macroscale using advanced computational simulation methods. Using state-of-the-art techniques I investigate a broad range of topics of current interest in the field of material science and engineering.

Publications

- 2020 B. Akgenc "Intriguing of two-dimensional Janus surface-functionalized MXenes: An ab initio calculation", Computational Materials Science, 171, 109231.
- 2019 A. Bafekry, C. Stampfl, B. Akgenc, M. Ghergherehchi "Control of C₃N₄ and C₄N₃ carbon nitride nanosheets' electronic and magnetic properties through embedded atoms", Physical Chemistry Chemical Physics, DOI:10.1039/C9CP06031F
- 2019 A. Bafekry, B. Akgenc, S. Farjami Shayesteh, B. Mortazavi "Tunable electronic and magnetic properties of graphene/carbon-nitride van der Waals heterostructures", Applied Surface Science, 144450.
- 2019 B. Akgenc "Two-dimensional Ti₂C monolayer (MXene) : Surface Functionalization-Induced Metal-Semiconductor Transition", Turkish Journal of Physics, 43 (5), 531-539.
- 2019 B. Akgenc "New predicted two-dimensional MXenes and their structural, electronic and lattice dynamical properties", Solid State Communications, 303, 113739.
- 2019 B. Akgenc "Two-dimensional black arsenic for Li-ion battery applications: a DFT study", Journal of Materials Science, 54, 13, pp 9543:9552.
- 2018 B.Akgenc, A.Kinaci, C. Tasseven and T. Cagin, " First principles calculations on stability and mechanical properties of various ABO₃ and their alloys", Materials Chemistry and Physics, 205, pp 315:324.
- 2018 B Akgenc, C Tasseven and T. Cagin, "Piezoelectricity in the AxA'(1-x)BO₃ and ABxB'(1-x)O₃ Ceramic Alloys", Materials Science Forum, 915, 34:38.
- 2018 B.Akgenc and T. Cagin, " Density functional theory of cubic zirconia and 6 mol - 15 mol doped yttria-stabilized zirconia: structural and mechanical properties", Turkish Journal of Physics
- 2016 S.D.Gunay, B. Akgenc and C. Tasseven "Modelling Superionic Behaviour of Plutonium Dioxide", High Temp. Mater. and Process, 35, 10, pp 999:1004.
- 2015 B Akgenc, C Tasseven, T. Cagin, "Stability and Mechanical Properties of AxA(1-x)'ByB(1-y)'O₃ Ceramics", TMS Middle East-Mediterranean Materials Congress on Energy and Infrastructure Systems, 423:432
- 2013 T. Wunderlich, B. Akgenc, U.Eckern, C.Schuster and U. Schwingenschlogl, "Modified Li chains as atomic switches", Sci. Rep. 3, 2605.
- 2013 B. Akgenc, U.Akdere, S.D. Gunay and C. Tasseven", NPT Simulation and Hypernetted-Chain Calculations of SrCl₂", 3rd AIP Conference Proceedings 1569 15:18
- 2013 S.D Gunay, B. Akgenc, U. Akdere and C. Tasseven, "Thermophysical Properties of alpha-Pu₂O₃: A New Potential Model", 3rd AIP Conference Proceedings 1569, 208:211.

Further publications

Submitted publications with peer review process

- 2019 A. Bafekry, B. Akgenc, M. Yagmurcukardes, "Stabilization of Monolayer ZnSb via Surface Functionalization: Metallic to Semiconducting Phase Transition", Journal of Materials Chemistry C
- 2019 A. Bafekry, B. Akgenc, M. Yagmurcukardes, B. Mortazavi "Formation of Bilayer and Trilayer van der Waals heterostructures of MoS₂/Janus MoSSe on graphitic boron-carbide and carbon-nitride monolayers", Applied Surface Science
- 2019 A. Bafekry, M. Nakhaee, B. Akgenc, F. M. Peeters, "Tight-binding model for Double Dirac nodal lines in Cu₂Si monolayer", IOP Nanotechnology, (Under Review)
- 2019 A. Bafekry, C. Stampfl, B. Akgenc, C. V. Nguyen "Embedding atoms into C₆N₈ and C₆N₈ carbon nitride monolayers with tunable electronic properties", Journal of Materials Chemistry C, (Under Review)
- 2019 B. Akgenc, A. Mogulkoc, E. Durgun "Phase-dependent electronic and magnetic properties of Ti₂C monolayers", Journal of Applied Physics, (Under Review)

Conference Presentations

- 2019 B. Akgenc, "Simetrik/asimetrik yuzey modifikasyonlari 2-boyutlu MXene'lerin piezoelektrik ozellikleri uzerine etkisi", 25. Yogun Madde Fizigi Ankara Toplantisi, Hacettepe Universitesi, 20 Aralik 2019, Ankara, TURKIYE
- 2019 B. Akgenc, A. Mogulkoc, E. Durgun "Phase Dependent Electronic and Magnetic Properties of Two-Dimensional Ti₂C (MXene) by First-Principle Calculations", 15th Nanoscience and Nanotechnology Conference (NanoTR-15), 3-6 November 2019, Lara-Antalya-TURKEY
- 2019 B. Akgenc, "Janus Monolayers of MXenes: A Dft Study", 35th International Physics Congress of Turkish Physical Society (TFD-35), 4-8 September 2019, Bodrum-Mugla-TURKEY.
- 2019 B. Akgenc, "High-Throughput Computational Survey Of Ordering Configurations In Pristine Mxene Alloys", International Natural Science, Engineering and Materials Technology Conference (NEM-2019), 9-10 September 2019, Istanbul-TURKEY.
- 2019 B. Akgenc "2-boyutlu MXene'lerin, fonon dinamik kararliginin, yapisal ve elektronik ozelliklerinin ilk prensipler yontemi ile arastirilmesi", VIII. Yogun Madde Fizigi Izmir Toplantisi, Izmir Yuksek Teknoloji Enstitusu 19 Nisan 2019, Izmir, TURKEY
- 2018 B. Akgenc, F. Iyikanat and H. Sahin, "First-principle calculations of Li-decorated black-Arsenic", 14th Nanoscience and Nanotechnology Conference, Izmir-TURKEY.
- 2018 B. Akgenc, "The effect of Y-doping on ZrO₂ : First-principle calculations ", 3rd International Organic Electronic Material Technology Conference, Igneada, TURKEY.
- 2017 B. Akgenc and A. Tekin, "Theoretical insights on mechanism for oxygen reduction reaction at S-N dual-doped graphene", 33rd International Physics Congress, Bodrum, TURKEY.
- 2017 B. Akgenc and T. Cagin, "First-principle calculation of oxygen diffusion in yttria-stabilized zirconia", 33rd International Physics Congress, Bodrum, TURKEY.
- 2017 B. Akgenc, C. Tasseven and T Cagin, "Piezoelectricity in the Ax_xA'(1-x)BO₃ and AB_xB'(1-x)O₃ Ceramic Alloys", The Tenth Japanese-Mediterranean Workshop on Applied Electromagnetic Engineering for Magnetic, Superconducting, Multifunctional and Nanomaterials, Izmir, TURKEY.
- 2016 B. Akgenc, C. Tasseven and T Cagin, "Piezoelectricity as smart materials", 2nd International Conference on Organic Electronic Material Technologies, Canakkale, TURKEY.
- 2015 B. Akgenc, C. Tasseven and T Cagin, " First principles investigations of structural, elastic, dielectric and piezoelectric properties of Ba,Sr,PbTiO₃ , Ba,Sr,PbZrO₃ and Ba,Sr,PbTi,ZrO₃ ceramics" APS March Meeting , San Antonio, USA.
- 2015 B. Akgenc, C. Tasseven and T Cagin, " Electro-Thermo-Mechanical and defect kinetics in AXA(1-x)ByB(1-y)O₃AXA(1-x)ByB(1-y)O₃" Canadian-American-Mexican Graduate Student Physics Conference, Oaxaca, MEXICO.
- 2015 B. Akgenc, C. Tasseven and T Cagin, "Lattice, elastic and piezoelectric properties in tetragonal BaTiO₃/PbTiO₃ first principle calculation", 2nd Texas AM Energy Institute Research Workshop, College Station, USA.
- 2015 B. Akgenc, C. Tasseven and T Cagin, "Stability, mechanical, dielectric and piezoelectric properties of Ax_xA'(1-x) ByB'(1-y)O₃ ", TMS Middle East, Mediterranean Materials Congress on Energy and Infrastructure System, Doha, QATAR
- 2014 B. Akgenc, C. Tasseven and T. Cagin , "DFT-based ab-initio studies of lattice, elastic and piezoelectric properties of Barium Titanate Polymorphs" International Institute for Multifunctional Materials for Energy Conversion Annual Meeting, College Station, USA.
- 2012 B. Akgenc, S.D. Gunay and C. Tasseven, "Temperature dependence of ionic diffusion of Thorium Dioxide", 29th International Physics Congress, Bodrum, TURKEY
- 2012 B. Akgenc, S.D. Gunay and C. Tasseven, "Classical Molecular Dynamics Simulation of PuO₂" Jubilee 20. Conference on Materials and Technology, Portoroz, SLOVENIA.
- 2012 C. Tasseven, B. Akgenc, S.D. Gunay and U. Akdere "Liquid state theory of plutonium dioxide: hypernetted chain approximation", 29th International Physics Congress, Bodrum, TURKEY.
- 2012 S.D. Gunay, B. Akgenc, U. Akdere and C. Tasseven, "Thermophysical Properties of alpha-Pu₂O₃: A New Potential Model", 3rd International Advances in Applied Physics and Materials Science Congress, Antalya-TURKEY

- 2012 S.D. Gunay, B. Akgenc, U. Akdere and C.Tasseven, "Studying superionic phase transition of plutonium dioxide from molecular dynamics simulation" 29th International Physics Congress, Bodrum, TURKEY
- 2012 B. Akgenc, U. Akdere, S.D. Gunay and C. Tasseven, "NPT Simulation and Hypernetted-Chain Calculation of SrCl₂", 3rd International Advances in Applied Physics and Materials Science Congress, Antalya-TURKEY
- 2010 T. Wunderlich, B. Akgenc, C. Schuster, U. Eckern, "Transmission through metallic chains: Role of distortions and contact geometry" , 21st Edgar-Luscher-Seminar,Klosters, SWITZERLAND
- 2009 B. Akgenc, K. Esmer, E. Tarcın, D. Avci and Y. Atalay, "Theoretical studies of molecular structure and vibrational spectra of 4-nitro-2-E-[2-(piperidin1yl)ethyl] iminomethylphenol", 26th International Physics Congress , Bodrum, TURKEY.